

$$\bullet \frac{d}{dx}(c)=0, \text{ where } c \text{ is constant}$$

$$\left\{ \begin{array}{l} \bullet \frac{d}{dx} \sin x = \cos x \\ \bullet \frac{d}{dx} \cos x = -\sin x \end{array} \right.$$

$$\left\{ \begin{array}{l} \bullet \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}} \\ \bullet \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}} \end{array} \right.$$

$$\left\{ \begin{array}{l} \bullet \frac{d}{dx} a^x = a^x \ln a \\ \bullet \frac{d}{dx} e^x = e^x \end{array} \right.$$

$$\left\{ \begin{array}{l} \bullet \frac{d}{dx} \sinh x = \cosh x \\ \bullet \frac{d}{dx} \cosh x = \sinh x \end{array} \right.$$

$$\left\{ \begin{array}{l} \bullet \frac{d}{dx} \sinh^{-1} x = \frac{1}{\sqrt{x^2+1}} \\ \bullet \frac{d}{dx} \cosh^{-1} x = \frac{1}{\sqrt{x^2-1}} \end{array} \right.$$

$$\bullet \frac{d}{dx}(x^n) = nx^{n-1}$$

$$\bullet \frac{d}{dx} \tan x = \sec^2 x$$

$$\bullet \frac{d}{dx} \cot x = -\csc^2 x$$

$$\bullet \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\bullet \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

$$\bullet \frac{d}{dx} \log_a x = \frac{1}{x \ln a}$$

$$\bullet \frac{d}{dx} \ln x = \frac{1}{x}$$

$$\bullet \frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$

$$\bullet \frac{d}{dx} \coth x = -\operatorname{csch}^2 x$$

$$\bullet \frac{d}{dx} \tanh^{-1} x = \frac{1}{1-x^2}$$

$$\bullet \frac{d}{dx} \coth^{-1} x = \frac{1}{1-x^2}$$

$$\bullet \frac{d}{dx} [f(x)]^n = n[f(x)]^{n-1} \frac{d}{dx} [f(x)]$$

$$\bullet \frac{d}{dx} \csc x = -\csc x \cot x$$

$$\bullet \frac{d}{dx} \sec x = \sec x \tan x$$

$$\bullet \frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\bullet \frac{d}{dx} \csc^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$$

$$\bullet \frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x$$

$$\bullet \frac{d}{dx} \operatorname{csch} x = -\operatorname{csch} x \coth x$$

$$\bullet \frac{d}{dx} \operatorname{sech}^{-1} x = \frac{-1}{x\sqrt{1-x^2}}$$

$$\bullet \frac{d}{dx} \operatorname{csch}^{-1} x = \frac{-1}{x\sqrt{1+x^2}}$$

Some Standard nth Derivative

$$\bullet \frac{d^n}{dx^n} (ax+b)^m = \frac{m!}{(m-n)!} a^n (ax+b)^{m-n} \text{ if } m \geq n$$

$$\bullet \frac{d^n}{dx^n} [\ln(ax+b)] = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}$$

$$\bullet \frac{d^n}{dx^n} \sin(ax+b) = a^n \sin\left(ax+b+n \cdot \frac{\pi}{2}\right)$$

$$\bullet \frac{d^n}{dx^n} e^{ax} \sin(bx+c) = (a^2+b^2)^{\frac{n}{2}} e^{ax} \sin\left(bx+c+n \tan^{-1} \frac{b}{a}\right)$$

$$\bullet \frac{d^n}{dx^n} e^{ax} \cos(bx+c) = (a^2+b^2)^{\frac{n}{2}} e^{ax} \cos\left(bx+c+n \tan^{-1} \frac{b}{a}\right)$$

$$\bullet \frac{d^n}{dx^n} \left(\frac{1}{ax+b}\right) = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$\bullet \frac{d^n}{dx^n} e^{ax} = a^n e^{ax}$$

$$\bullet \frac{d^n}{dx^n} \cos(ax+b) = a^n \cos\left(ax+b+n \cdot \frac{\pi}{2}\right)$$

Leibniz's Theorem

$$\bullet \frac{d^n}{dx^n} (uv) = {}^nC_0 u^{(n)} v + {}^nC_1 u^{(n-1)} v' + {}^nC_2 u^{(n-2)} v'' + \dots + {}^nC_{n-1} u' v^{n-1} + {}^nC_n u v^n$$

***Differentiation of
Trigonometric Functions***

- 1) $\frac{d}{dx}(\text{Sin}x) = \text{Cos}x \frac{d}{dx}(x)$
- 2) $\frac{d}{dx}(\text{Cos}x) = -\text{Sin}x \frac{d}{dx}(x)$
- 3) $\frac{d}{dx}(\text{tan}x) = \text{Sec}^2x \frac{d}{dx}(x)$
- 4) $\frac{d}{dx}(\text{Cot}x) = -\text{Cosec}^2x \frac{d}{dx}(x)$
- 5) $\frac{d}{dx}(\text{Sec}x) = \text{Sec}x.\text{tan}x \frac{d}{dx}(x)$
- 6) $\frac{d}{dx}(\text{Cesec}x) = -\text{Cosec}x.\text{Cot}x \frac{d}{dx}(x)$

***Differentiation of
Hyperbolic Functions***

- 1) $\frac{d}{dx}(\text{Sinh}x) = \text{Cosh}x \frac{d}{dx}(x)$
- 2) $\frac{d}{dx}(\text{Cosh}x) = \text{Sinh}x \frac{d}{dx}(x)$
- 3) $\frac{d}{dx}(\text{tanh}x) = \text{Sech}^2x \frac{d}{dx}(x)$
- 4) $\frac{d}{dx}(\text{Coth}x) = -\text{Cosech}^2x \frac{d}{dx}(x)$
- 5) $\frac{d}{dx}(\text{Sech}x) = -\text{Sech}x.\text{tanh}x \frac{d}{dx}(x)$
- 6) $\frac{d}{dx}(\text{Cosech}x) = -\text{Cosech}x.\text{coth}x \frac{d}{dx}(x)$

***Differentiation of Inverse
Trigonometric Functions***

- 1) $\frac{d}{dx}(\text{Sin}^{-1}x) = \frac{1}{\sqrt{1-x^2}} \frac{d}{dx}(x)$
- 2) $\frac{d}{dx}(\text{Cos}^{-1}x) = \frac{-1}{\sqrt{1-x^2}} \frac{d}{dx}(x)$
- 3) $\frac{d}{dx}(\text{tan}^{-1}x) = \frac{1}{1+x^2} \frac{d}{dx}(x)$
- 4) $\frac{d}{dx}(\text{Cot}^{-1}x) = \frac{-1}{1+x^2} \frac{d}{dx}(x)$
- 5) $\frac{d}{dx}(\text{Sec}^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}} \frac{d}{dx}(x)$
- 6) $\frac{d}{dx}(\text{Cosec}^{-1}x) = \frac{-1}{|x|\sqrt{x^2-1}} \frac{d}{dx}(x)$

***Differentiation of Inverse
Hyperbolic Functions***

- 1) $\frac{d}{dx}(\text{Sinh}^{-1}x) = \frac{1}{\sqrt{1+x^2}} \frac{d}{dx}(x)$
- 2) $\frac{d}{dx}(\text{Cosh}^{-1}x) = \frac{1}{\sqrt{x^2-1}} \frac{d}{dx}(x)$
- 3) $\frac{d}{dx}(\text{tanh}^{-1}x) = \frac{1}{1-x^2} \frac{d}{dx}(x)$
- 4) $\frac{d}{dx}(\text{Coth}^{-1}x) = \frac{-1}{x^2-1} \frac{d}{dx}(x)$
- 5) $\frac{d}{dx}(\text{Sech}^{-1}x) = \frac{-1}{|x|\sqrt{1-x^2}} \frac{d}{dx}(x)$
- 6) $\frac{d}{dx}(\text{Cosech}^{-1}x) = \frac{-1}{|x|\sqrt{x^2+1}} \frac{d}{dx}(x)$

Integration Formulas

- 1 $\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C$
- 2 $\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C$
- 3 $\int \tan(ax + b) dx = -\frac{1}{a} \ln |\cos(ax + b)| + C = \frac{1}{a} \ln |\sec(ax + b)| + C$
- 4 $\int \cot(ax + b) dx = \frac{1}{a} \ln |\sin(ax + b)| + C$
- 5 $\int \operatorname{cosec} x dx = \ln |\operatorname{cosec} x - \cot x| + C = \ln \left| \tan \frac{x}{2} \right| + C$
- 6 $\int \sec x dx = \ln |\sec x + \tan x| + C = \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C$
- 7 $\int e^{ax+b} dx = \frac{1}{a} \times e^{ax+b} + C$
- 8 $\int a^{bx+c} dx = \frac{1}{b \ln a} \times a^{bx+c} + C$
- 9 $\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \sinh^{-1} \left(\frac{x}{a} \right)$
- 10 $\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \cosh^{-1} \left(\frac{x}{a} \right)$
- 11 $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$
- 12 $\int \frac{1}{\sqrt{x^2 + a^2}} dx = \sinh^{-1} \left(\frac{x}{a} \right) + C = \ln \left(\frac{x + \sqrt{x^2 + a^2}}{a} \right) + C$
- 13 $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \cosh^{-1} \left(\frac{x}{a} \right) + C = \ln \left(\frac{x + \sqrt{x^2 - a^2}}{a} \right) + C$
- 14 $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$
- 15 $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + C$
- 16 $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$
- 17 $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$